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A Comparison of Layered Tetrahedral and Cartesian meshing in Additive Manufacturing Simulation

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Abstract

In additive manufacturing simulation most of the time the overall aim is to compensate residual stress induced deformations of a printed part. For this goal a high result accuracy is needed. In order to achieve this, simulation engineers have to take numerous decisions. This article provides assistance in the determination of a reasonable meshing type using a complex to mesh test sample that is close to the real geometry of round metallic fatigue test samples. The traditional Cartesian meshing strategy with and without the use of a projection factor is compared with layered tetrahedral meshing, which is newly introduced to additive manufacturing simulation. Additionally the displacement results are set side by side with a scan of an additive manufactured part. The study shows that layered tetrahedral meshing can achieve more accurate results with coarser mesh sizes, especially in regions of high curvature.

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1. Introduction

Additive Manufacturing (AM) has become increasingly popular in recent years. Especially for metal parts with powder bed fusion processes. This is reflected by more than a 40% increase in the sales volume of metal powder produced for the additive manufacturing industry [1]. The benefits of AM, opposed to traditional manufacturing technologies, are in the possibility to create parts with a high level of complexity quite easy, without the need for casting moulds or tooling. Despite the possible highly complex shapes, one has to deal with manifold and complex physical effects during the additive manufacturing process. This leads to the problem that not yet every single effect is well understood [2, 3]. One of the effects difficult to predict is the thermally induced residual stresses and therefrom resulting deformations [4]. These residual stresses lead to large distortions of the part during production and the need for several attempts in support structure design before a part can be printed without failure. As this is highly time consuming and also expensive, simulations are used to predict the deformations during the print. With those simulation results the part or the support structure can be adapted in order to be printed without failure at the first attempt. Because of the large number of machine parameters, physical effects that are present in an additive manufacturing process and limited computational resources, it is necessary to restrict the simulated effects. It is important to compare the results to corresponding experiments [5]. For fast computation and accurate results, correct meshing is very important.

The aim of this paper is to provide a tool for the selection of a suitable meshing strategy, which is capable of generating accurate results with minimal computational effort. Therefore this article points out the differences between the, for additive manufacturing simulations, newly introduced layered tetrahedral and the traditional Cartesian meshing strategy. An artificial test sample is designed to have a complex to mesh geometry and is then simulated with different mesh types and element sizes until convergence is reached. The simulation models are then compared with each other regarding computation time, element quality and result accuracy as well

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as a structural error energy. Finally the results of the layered tetrahedral mesh are compared with a 3D scan of an actual printed part.

2. State of the Art

2.1. Laser Powder Bed Fusion

Laser Powder Bed Fusion (L-PBF) commonly referred as Selective Laser Melting (SLM) or Direct Metal Laser Sintering (DMLS) is a layered manufacturing process in the field of AM. It is one of the most commercialized metal AM processes. As opposed to subtractive manufacturing methods, metal powder is molten, under the use of a laser beam, successively layer upon layer to generate the part [6]. The process starts with a thin layer of powder on the build plate. After melting a layer, the build plate is lowered by a specific amount, specifying the layer height, and new powder is placed on top by a recoating device. The scanning path of the laser is controlled by a file containing vectors for each layer. This file is generated beforehand by slicing a 3D model, which is usually created with a Computer Aided Design (CAD) system. In order to extinguish the effect of oxidation in the build chamber, remaining oxygen is evacuated by flooding the chamber with inert gas. Typically the layer height ranges from 20 to 50 µm. A schematic of the process is shown in Fig. 1.

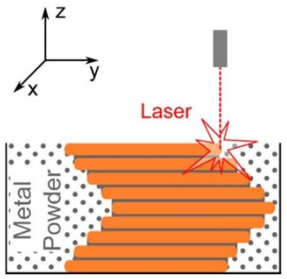


Fig. 1. Schematic of the L-PBF process.

2.2. Additive Manufacturing Simulation

The Finite-Element-Method (FEM) has become the most popular mesh-based numerical method for the calculation of structural and thermal models and therefore also for physical phenomena as residual stresses and distortions during the AM process. The main idea behind the FEM is that a complex geometry can be discretized in a finite number of much simpler objects, the elements. Those elements form a mesh representing the actual geometry reduced to a finite number of unknowns. The finite number of unknowns leads to a finite number of equations to be solved, either in an explicit or implicit way. Explicit methods can run into numerical instability, if the time step is too large. However implicit solutions might fail to converge for heavily non-linear calculations [7].

Simulations of the additive manufacturing process are split into two separate systems. A thermal and a mechanical simulation. These simulations can run simultaneously as coupled calculation or in sequence as uncoupled calculations. Coupled simulations are more accurate, but at the same time more computational expensive. As input for both systems material properties, the parts geometry, which later has to be meshed, as well as initial and boundary conditions have to be specified. During the solution process the temperature field at each layer is calculated. To reduce computational cost, the heat for the phase change of the powder can be applied at once for each layer [8]. In this way there is no need for a detailed simulation of the scanning vectors. Based on the thermal results, the displacements of the mechanical model are evaluated. In order to calculate the model with an increasing number of layers, the elements of each layer are suppressed and then reactivated at the right time using the element birth and death approach. Due to the transient analysis in layers, the mesh also needs to be in form of layers. At the moment there are two different meshing options, which will be described in the next sections of this article.

2.3. Cartesian meshing

Cartesian or voxel meshing is the most commonly used mesh type for current additive manufacturing simulations [9]. The mesh consists of cubes (voxels) with a, for the whole model consistent, specified size. This leads to a layered mesh, which is necessary to mimic the part generation in layers like in powder bed fusion processes using element birth and death. Due to the voxel elements with identical edge length on each side of the cube, the representation of the geometry is only an approximation. Depending on the size of the elements, tiny features are either not expressed in detail, leading to an inaccurate simulation, or the computational cost of the simulation model highly increases. In order to cope with the inaccurate representation of the geometry using hexahedron elements, ANSYS® uses a so called projection factor. This factor ranges from zero to one and describes the level of representation with zero as no adaption of the elements. With an increasing value of the factor, the elements get more and more distorted to fit the actual geometry of the part. The resulting skewness and uneven aspect ratio leads to a worse element quality. Nevertheless more accurate simulation results are possible using this meshing method. Cartesian meshes can use linear, quadratic or even higher polynomial approaches. The hexahedron shaped elements then have eight (linear) or twenty (quadratic) nodes per element.

2.4. Layered tetrahedral meshing

Layered tetrahedral meshing was first introduced for additive manufacturing simulation in 2019 by ANSYS®. Its aim is to simplify the representation of the small detailed geometry of AM parts without increasing the level of complexity of the model [10]. The tetrahedral shape of the elements allows the meshing algorithm to achieve a better fitting mesh with only small element distortions. Despite the higher number of elements in the simulation model using this meshing method, accurate results can be achieved in less time than with Cartesian meshes [9]. Layered tetrahedral meshing differs from traditional tetrahedral meshing in terms of element orientation. Within the meshing parameters, the layer height is defined, at which all elements need to form a flat surface. The

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layer height can, but does not have to, be equal with the element size, as elements are allowed to be smaller. The mesh is generated from one part surface on with evenly thick slices consisting of tetrahedrons for element birth and death.

The tetrahedron shaped elements of a layered tetrahedral mesh commonly have five (linear approach) or ten (quadratic approach) nodes per element. As there are less nodes per element than with the hexahedron shaped ones, more elements are required to reach the same result accuracy.

3. Implementation

3.1. Selection of the test geometry

The geometry selected for this simulations is almost identical with metal fatigue test samples described in ASTM E466-15 [11] and similar to round metallic tensile test samples described in the standard ASTM E8-04 [12]. It is defined by a diameter of 10 mm at the top and the bottom of the rod, reaching 15 mm into the part. The middle section is, opposed to regular tensile test samples designed using a constant radius of 46.25 mm, leaving a diameter of 5 mm at the smallest cross-section. The total height of the rod is 60 mm.

This design was selected, because of its close similarity to tensile test samples and the rather small height for shorter simulation time. Also the constant radius of the middle section from side view is better for this experiment, because it is more complex to mesh. For the same reason a cylindrical test sample was preferred over rectangular shapes.

3.2. Modelling & Simulation

The previously described test geometry is placed directly on a round plate with 100 mm diameter and 20 mm thickness, representing the build plate. Furthermore it is oriented in an upright position on the plate without the use of any support. As material for the build plate and the part the ANSYS® additive material library titanium alloy Ti-6Al-4V is used with temperature dependent properties for density, specific heat, thermal conductivity and isotropic elasticity as well as bilinear isotropic hardening. The melting temperature of the titanium alloy is set to 1605 °C. In the next step twelve variants are created. For each meshing type available (Cartesian, Cartesian with projection factor and layered tetrahedral) variants with a specified element size of 1 mm, 0.75 mm, 0.5 mm and 0.25 mm are created. Further in this work the mesh size of 0.25 mm is not considered for the results as the computational cost with computation times of more than 96 hours exceeded the available ressources. For layered tetrahedral meshing the layer thickness is set to be equivalent to the element size resulting in one element in z-direction per layer. The projection factor for the Cartesian mesh with use of the projection factor is set to 0.5 of 1 as a maximum. All other parameters in the meshing options are left on their default value. The differences between the mesh types and how accurate the geometry can be represented is shown with a 1 mm element size in figure 2. The elements themselve follow a quadratic approach and have therefore 10 nodes for tetrahedrons and 20 nodes for hexahedrons. The mesh for the base plate is a hexahedron

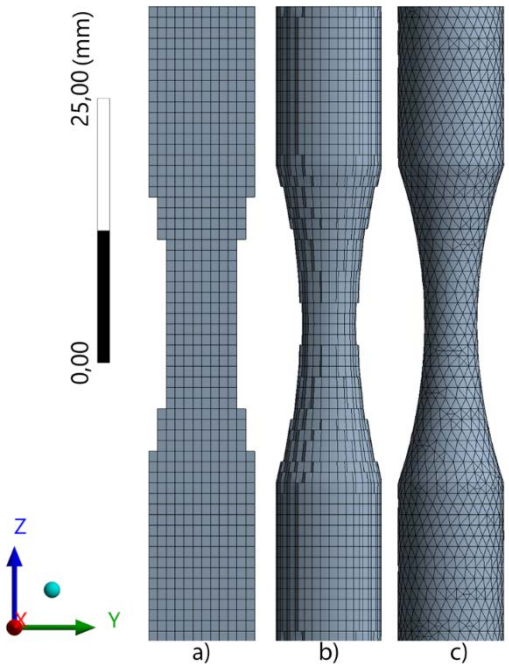


Fig. 2. Mesh types with size 1 mm: Cartesian (a), Cartesian with projection factor (b) and layered tetrahedral (c) mesh.

dominant mesh with an element size of 10 mm as it is not of a high importance for this calcualtion.

In the AM process settings, all variables for the virtual additive manufacturing are set. As manufacturing process, powder bed fusion is selected. Furthermore the layer height is set to 20 µm, hatch spacing to 80 µm and scanning speed to 600 mm/s. The time between layers is set to be 12 seconds. As the simulation is only for a single part on the build plate, no part multiplier is defined and the number of heat sources is set to one. Additional parameters are grouped in those for during the build and some for the cool down phase. In these two groups only the preheat temperature and room temperature are set with an equal value of 24 °C. All other parameters for additive manufacturing such as the powder and gas convection coefficients are left on their default value.

This simulation is constructed as uncoupled calculations like it is described in section 2.2 of this article. This means a transient thermal analysis is followed by a static structural analysis. For the thermal analysis two load steps are created. One for the build phase and one for cool down. For each step one temperature boundary condition with 24 °C is set on the lower surface of the build plate. For the structural analysis there are as well two load steps for build and cool down. The boundary condition for both of them defines the fixation of all degrees of freedom for the lower surface of the build plate. The connection between the build plate and the part to be printed is realized using a bonded contact. Because of the non-linear material behaviour both simulations run as non-linear calculations. All results are computed on a workstation with 64 GB of RAM, Intel® Core™ i7 CPU with up to 4.3 GHz and a Nvidia® GeForce® GTX 1080 Ti for forced GPU acceleration.

3.3. Fabrication of the test sample

In the pre-processing, the part shown in figure 3 was placed in the centre of the build plate without the use of any support and

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Fig. 3. TiAl6V4 test sample produced using the L-PBF process.

was then sliced into 3,000 consecutive layers using the machine manufacturers build processor. No support structure was used to eliminate the effect of it on residual stresses and limit the results to the actual part. The selected layer height was 20 µm because of good experiences from previous prints. The test sample was fabricated according to the parameters of the simulation on a Trumpf TruPrint 1000 Multilaser selective laser melting machine using a single laser. The used material was TiAl6V4 (Grade 23) powder obtained from SLM® Solutions with a particle size of 20 – 63 µm. Because of the small proportion of larger sized particles it is conditionally applicable for a layer height of 20 µm, but still delivers good results and was used due to availability. As in the simulation only the melting temperature is specified, the laser power that matches this melting temperature had to be estimated based on the energy input using empirical values from previous prints. The selected power was 83 W for a scanning speed of 600 mm/s. The time between layers was set to 12 seconds to match the simulation. Another parameter that has to be set on the L-PBF machine and which is not present in simulations is the air speed of the inert gas. In this case it was set to 2.5 m/s, resulting in an inert gas flow of 4.4 l/min. In order to assure a non-oxidizing build environment, the maximum level of residual oxygen in the build chamber was allowed to be 0.01%. At the beginning of the print the temperature inside the build chamber was 24.5 °C with a relative humidity of 40%.

4. Results & Analysis

4.1. Differences in the mesh types

The simulation results for the structural error energy (SERR) and the displacement in z-direction are shown in figure 4 and 5. These results are only two of the result values and model properties that were used to compare the three mesh types. An overview of the values for each mesh type and mesh size is listed in Table 1. Starting with the number of nodes it can be seen that for each mesh size the layered tetrahedron mesh has up to almost double as much nodes as the Cartesian meshes. For the number of elements, this difference is even bigger. This leads to a higher number of total degrees of freedom in the model and therefore to more equations to be solved. With more equations the computational cost and the calculation time increases. The shortest time (~8h) for a mesh size of 0.5 mm was reached by the Cartesian mesh with the projection factor. Often the aspect ratio, skewness and minimum edge angle of the elements are reason for failed simulations because of elements turning inside out or equations that fail to converge. Therefore the element quality is a significant property to look at. Mesh statistics show, as

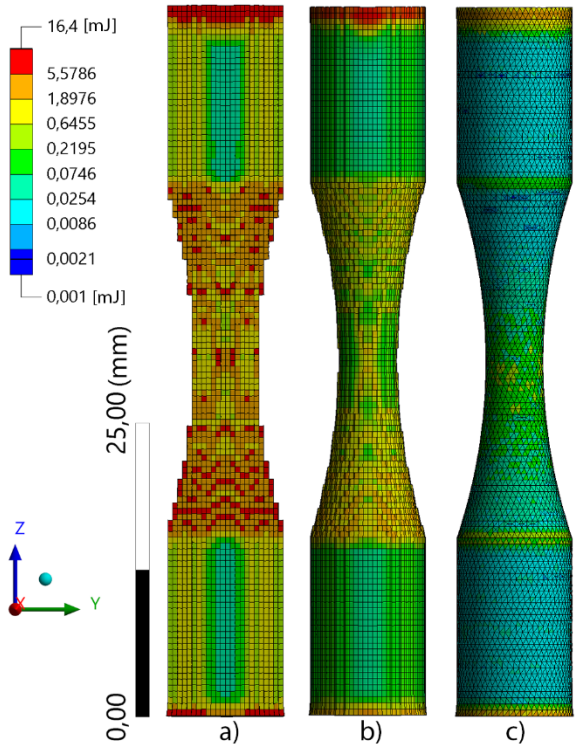


Fig. 4. Structural error energy in mJ on a logarithmic scale for Cartesian (a), Cartesian with projection factor (b) and layered tetrahedral (c) meshing with size 0.5 mm.

Table 1. Comparison of result values for different mesh types with best results marked in bold.

	Cartesian			Cartesian with projection			Layered tetrahedral		
Mesh size (mm)	1	0.75	0.5	1	0.75	0.5	1	0.75	0.5
Number of nodes	31,244	45,419	208,855	30,132	44,291	204,651	42,260	81,646	294,015
Number of elements	6484	9588	47,124	6,222	9,324	46,107	27,515	53,102	207,857
Average quality	0.971	**0.994**	0.967	0.907	0.950	0.936	0.799	0.716	0.818
Computation time (s)	1,170	2,668	30,722	1,094	2,377	**29,214**	1,550	3,481	38,288
Average Structural error; SERR (mJ)	41.872	18.843	2.106	37.622	16.194	1.792	6.162	2.664	**0.302**
Characteristic element length (mm)	0.839	0.741	0.404	0.713	0.671	0.380	0.839	0.631	0.442

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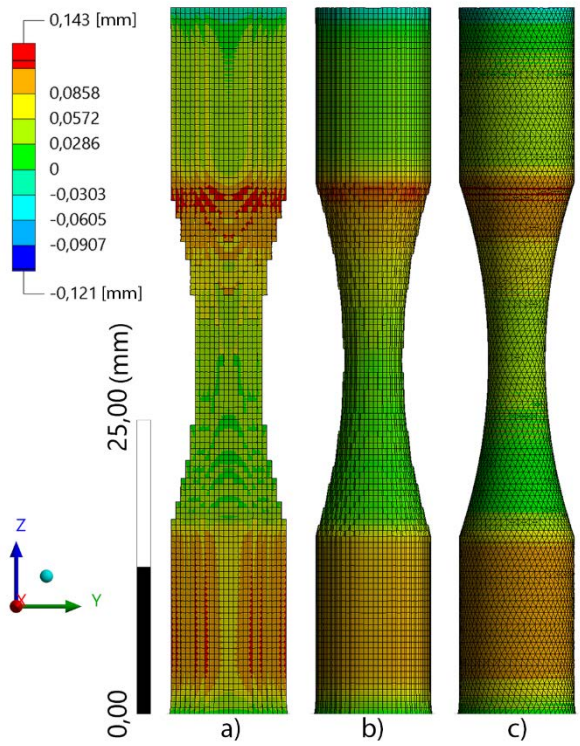


Fig. 5. Displacement in z-direction in mm for Cartesian (a), Cartesian with projection factor (b) and layered tetrahedral (c) meshing with a size of 0.5 mm.

expected, the best value of the average element quality for the Cartesian mesh with 99.4%. Even the elements with distortion induced by the projection factor have still an average quality above 90%. This is also reflected by the low minimum element quality of the tetrahedral mesh with approximately 20%.

As the previous mentioned values and properties only give an overview on the simulation process itself, the results of figure 4 and 5 are used to compare the mesh types for result accuracy. In both colour plots can be seen that the inconsistent colour variations, especially for the Cartesian mesh, are dependent on geometry representation accuracy. The fast and inhomogeneous colour transitions are an indicator for inaccurate results. This can also be seen in figure 4 as it shows the estimated structural error energy (SERR) on a logarithmic scale with up to the order of two higher results for the Cartesian mesh. In fact the arithmetic mean value over all elements in the model is 0.3 mJ for the layered tetrahedral mesh compared to 2.1 mJ for the Cartesian. This difference in result accuracy is not changing with even smaller elements as all models reached result convergence. The layered tetrahedron mesh converged even already with a mesh size of 0.75 mm. As criteria for a converged model the gradient of the estimated error has to be of an order close to the squared characteristic element length following the example of [13] for displacement errors.

4.2. Comparison of the simulation and the print

The printed test sample was scanned using a Creaform Handyscan 700™ with an accuracy of up to 0.03 mm. Measurements on the scanned surface in z-direction showed a mean shrinkage at the centre point of the top surface of 0.36 mm. Compared to the results displayed in figure 5 the displacement in z-direction of the scanned part is in the same order but three times as high. Figure 6 views the geometric deviations between the deformed geometry from figure 5 (c) and the scanned part as colour plot. Both polygon models are pre-aligned using a best-fit algorithm. After the pre-alignment the correct starting point in z-direction is assured by adding a plane through the bottom surface and shifting the scan along the z-axis until it is coincident with the simulation base plate. The top surface of the actual sample is around 0.2 mm lower than the one from the simulation and is therefore coloured blue with a negative sign. It is worth noticing that the lower part of the radius is almost coincident in the scan and the simulation, whereas the upper part shows an increasing deviation starting from the centre point. The red color indicates a positive deviation from the simulation geometry, which means the diameter of the sample is larger than the simulation. This is consistent with a total shrinkage of the sample in z-direction. Deviations on the cylindrical parts of the test sample should be neglected as they are not constant in x- and y-direction and hence afflicted with inaccuracy in alignment of both z-axis.

4.3. Result discussion

The result accuracy shows a strong dependency on the geometry representation, especially seen in figure 4, at each point of the mesh. In other words, Cartesian meshing is still reasonable for models with less curvature. Nevertheless the tetrahedral mesh can be coarser and showed, other than the Cartesian meshes, also with 0.75 mm element size, accurate results. For almost the same accuracy the 0.75 mm tetrahedral mesh needed only 3,481s. The achieved time savings due to reduced computational cost are in the order of ten. The layered tetrahedral mesh shows a more geometrical accurate and therefore more realistic representation.

Beside the differences between the presented types of meshing, also the comparison to a real part is analysed. Measurements showed a qualitative agreement of the simulation with the printed sample. However quantitative

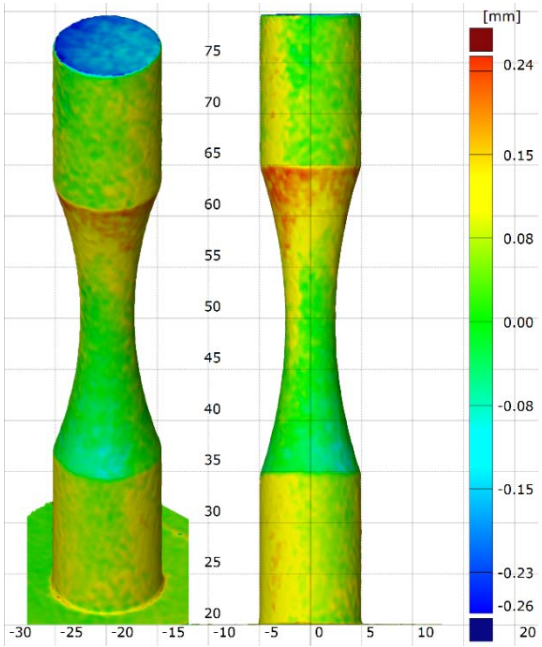


Fig. 6. Geometric deviation between the deformed tetrahedral mesh with 0.5 mm size and a scan of the test sample created using L-PBF

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differences between simulation and scan could be observed. Reasons for this deviation are most likely in the settings of the simulation. One of the biggest assumptions that is made is the energy input, as the melting temperature of the material is used instead of the laser power of the actual print. Another simplification is that heat is applied layer by layer and not elementwise following a hatch pattern. Also to reduce computational cost, the surrounding powder was not included as a mesh in these simulations. Available parameters to adjust the simulation are limited and for example the gas flow or the inconsistency of the build chamber temperature are not considered. On top of that, thermal properties such as convection coefficients, thermal conductivity or temperature dependent elasticity are often not available for the specific powder used for the L-PBF process. Hence materials from the software's standard library are used. Additional to the simplifications of the simulation, the used powder for this test sample was not optimal, because of its partial large grain size paired with the selected layer height. Larger particles are taken away by the recoating device, resulting in a lower density. Nevertheless the powder was used due to the reason of availability.

5. Conclusion

Traditional types as the Cartesian mesh with and without a projection factor and new approaches like the layered tetrahedral mesh were faced. The results show, that for the same mesh size, the layered tetrahedral mesh delivers a smoother result that fits the actual geometry of the part better especially in regions of high curvature. This does not necessarily lead to more accurate results as the point where stresses and displacements are evaluated is also important.

In general it is difficult to assure which mesh type suits generic cases the best, because of the dependency on the geometry. Also there are rather large deviations between the actual printed part and the simulations whilst the differences in displacements for all mesh types are quite small. Nevertheless it is shown that the best result convergence together with the smallest structural error energy is reached by the layered tetrahedral mesh even though the best element quality is achieved by the Cartesian mesh. It has to be pointed out that the overall results show deviations from each other using different meshing strategies for the same element size. Compared to the resolution of the 3D scanner those oscillations are way smaller and therefore not applicable for determination of the best mesh type. For the minimum displacement in z-direction the variations between the evaluated meshing types are even smaller than the surface roughness of the top surface. This means that the results of all three meshing strategies will show the same deviations if they are compared to the 3D scan.

In future works the results of this paper could be extended by the evaluation of simulated residual stresses and their validation using techniques such as neutron diffractometry. Also the applicability to other geometries with less curvature or more residual stress induced deformations can be evaluated. As the overall aim often is the compensation of these deformations with a pre-distorted model the benefits of the closer geometrical representation of layered tetrahedral meshes could be a topic for further investigations.

Using the same element size, the layered tetrahedral mesh has shown to be relatively computational expensive compared to the other two meshing strategies. However, if a small mesh size is not relevant, layered tetrahedral meshes can deliver results faster due to the result convergence with coarser meshes. Another benefit is a better more accurate representation of the parts deformations especially for complex geometries. For thin features and places with hypothetically high stress intensity factors, a layered tetrahedral mesh works best in order to minimize the possible discretization induced error.

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